Review.

1. Poisson distribution.

X = # of occurrences for a given interval or region.

$X \sim \text{Poisson}(\lambda)$.

λ = mean occurrences for that given interval or region.

$$f(x) = P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!} \qquad x = 0, 1, \ldots$$

If. I change the interval / region to be t.

Y = # of occurrences for t.

$Y \sim \text{Poisson}(\lambda t)$.

2. pdf, cdf.

pdf → cdf: $F(x) = \int_{-\infty}^{x} f(t)\,dt$.

cdf → pdf: $f(x) = \frac{d}{dx} F(x)$.

Method 2: $P(X<10) = P(X \le 10) = F(10) = 1 - e^{-\frac{1}{3}\times 10} = \underline{0.9643}$ ②

Method 3: Y = # calls in 10 minutes

$Y \sim \text{Poisson}\left(\frac{1}{3}\times 10 = \frac{10}{3}\right)$

$$P(Y \ge 1) = 1 - P(Y<1)$$

$$= 1 - P(Y=0)$$

$$= 1 - \frac{e^{-\frac{10}{3}}\left(\frac{10}{3}\right)^0}{0!} = \underline{0.9643}$$

c) Mean waiting time between calls?

X = waiting time.

$$E(X) = \frac{1}{1/3} = 3 \text{ min.}$$

d) Find 95th. percentile of the waiting time.

$$0.95 = \underline{P(X \le x_0)} = F(x_0) = 1 - e^{-\frac{x_0}{3}}$$

$$\Rightarrow \quad e^{-\frac{x_0}{3}} = 1 - 0.95 = 0.05$$

$$-\frac{x_0}{3} = \ln 0.05$$

$$x_0 = -3 \ln 0.05 = 8.99 \text{ min.}$$